

Shannon Entropy

math-foundations / concept 36

First, an example

Before the general theory, fix intuition with a coin. Let $X \sim \text{Bernoulli}(p)$, so $X = 1$ with probability p and $X = 0$ with probability $1 - p$. The Shannon entropy in bits is

$$H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p).$$

Read aloud: “H of p equals minus p log-base-two of p, minus one-minus-p log-base-two of one-minus-p.” At $p = 0.5$ the coin is fair and $H(0.5) = 1$ bit — the maximum, because we are maximally uncertain. At $p = 0.9$ the coin is biased and $H(0.9) \approx 0.469$ bits — less uncertain, since we can usually guess heads. At $p = 1$ the outcome is deterministic and $H(1) = 0$ bits — no surprise at all. Operationally, an optimal Huffman code for i.i.d. draws of X uses on average $H(p)$ bits per outcome, so entropy is the price (in bits) of describing one sample.

1 Motivation

Self-information $I(x) = -\log p(x)$ measures the surprise of a single outcome. Shannon entropy is the *expected* self-information: the average surprise we feel when drawing from p repeatedly. Equivalently, it is the optimal expected length (in bits, nats, hartleys, depending on the log base) of a prefix-free code for the random variable.

2 Definitions

Definition 1 (Shannon entropy, discrete). Let X be a discrete random variable on a countable alphabet $\mathcal{X}$ with probability mass function p . The *Shannon entropy* of X is

$$H(X) = \mathbb{E}[-\log p(X)] = - \sum_{x \in \mathcal{X}} p(x) \log p(x),$$

Read aloud: “H of X equals the expectation of minus log p of X, equals minus the sum over x in script-X of p of x times log p of x.” with the convention $0 \log 0 = 0$ (justified by $\lim_{p \rightarrow 0^+} p \log p = 0$).

Definition 2 (Differential entropy, continuous). Let X be a continuous random variable with density f . The *differential entropy* of X is

$$h(X) = - \int_{\mathbb{R}} f(x) \log f(x) dx,$$

Read aloud: “little-h of X equals minus the integral over the real line of f of x times log f of x, d x.” whenever the integral exists. Unlike discrete entropy, $h(X)$ may be negative and is not invariant under change of variables.

The base of the logarithm fixes the units: $\log_2 \Rightarrow$ bits; $\ln \Rightarrow$ nats; $\log_{10} \Rightarrow$ hartleys. We use $\log$ to denote a generic base unless specified.

3 Basic properties

Lemma 3 (Non-negativity). *For any discrete random variable X , $H(X) \geq 0$, with equality if and only if X is deterministic (i.e., some outcome has probability 1).*

Proof. Each summand $-p(x) \log p(x)$ is non-negative because $0 \leq p(x) \leq 1$ implies $\log p(x) \leq 0$. Equality in the sum requires every term to vanish, which forces $p(x) \in \{0, 1\}$ for all x . $\square$

4 Maximum-entropy theorem

The following result is the central inequality of this lesson.

Theorem 4 (Uniform maximises entropy). *Let X be a discrete random variable on a finite alphabet $\mathcal{X}$ with $|\mathcal{X}| = n$. Then*

$$H(X) \leq \log n,$$

Read aloud: “ H of X is less than or equal to $\log n$.” with equality if and only if X is uniform on $\mathcal{X}$ (i.e., $p(x) = 1/n$ for every $x \in \mathcal{X}$).

Proof. We use Jensen’s inequality applied to the strictly concave function $\log$. Let $\mathcal{X}^+ = \{x \in \mathcal{X} : p(x) > 0\}$ denote the support of p , and write $m = |\mathcal{X}^+| \leq n$. By definition,

$$H(X) = - \sum_{x \in \mathcal{X}^+} p(x) \log p(x) = \sum_{x \in \mathcal{X}^+} p(x) \log \frac{1}{p(x)}.$$

Read aloud: “ H of X equals minus the sum over x in the support of p of $x \log p$ of x , equals the sum over x in the support of p of x times $\log$ of one over p of x .” View the right-hand side as the expectation $\mathbb{E} \left[\log \frac{1}{p(X)} \right]$, where the expectation is taken under p . Since $\log$ is concave, Jensen’s inequality gives

$$\mathbb{E} \left[\log \frac{1}{p(X)} \right] \leq \log \left(\mathbb{E} \left[\frac{1}{p(X)} \right] \right).$$

Read aloud: “the expectation of $\log$ of one over p of X is less than or equal to $\log$ of the expectation of one over p of X .” Compute the inner expectation:

$$\mathbb{E} \left[\frac{1}{p(X)} \right] = \sum_{x \in \mathcal{X}^+} p(x) \cdot \frac{1}{p(x)} = \sum_{x \in \mathcal{X}^+} 1 = m.$$

Read aloud: “the expectation of one over p of X equals the sum over x in the support of p of x times one over p of x , equals the sum of ones over the support, equals m .” Therefore

$$H(X) \leq \log m \leq \log n,$$

Read aloud: “ H of X is less than or equal to $\log m$, which is less than or equal to $\log n$.” where the second inequality uses $m \leq n$ and monotonicity of $\log$.

For the equality condition: $\log$ is *strictly* concave, so Jensen’s inequality is strict unless $1/p(X)$ is constant on the support, which means $p(x) = 1/m$ for all $x \in \mathcal{X}^+$. Combined with $m \leq n$, equality $H(X) = \log n$ requires $m = n$ and p uniform on $\mathcal{X}$. Conversely, if p is uniform on $\mathcal{X}$, then $H(X) = - \sum_x (1/n) \log(1/n) = \log n$. $\square$

Corollary 5. *For any discrete random variable on n outcomes, $0 \leq H(X) \leq \log n$. The lower bound is achieved by deterministic X , and the upper bound by uniform X .*

5 Joint entropy and subadditivity

Definition 6 (Joint entropy). For discrete random variables X, Y with joint pmf $p(x, y)$,

$$H(X, Y) = - \sum_{x, y} p(x, y) \log p(x, y).$$

Read aloud: “H of X comma Y equals minus the double sum over x and y of p of x comma y times log p of x comma y.”

Theorem 7 (Subadditivity). $H(X, Y) \leq H(X) + H(Y)$, with equality iff X and Y are independent.

The proof uses non-negativity of KL divergence applied to the joint vs. product of marginals; we defer it to the cross-entropy / KL lesson.

6 Bernoulli entropy as a worked example

Example 8 (Binary entropy function). For $X \sim \text{Bernoulli}(p)$,

$$H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p).$$

Read aloud: “H of p equals minus p log-base-two of p, minus one-minus-p log-base-two of one-minus-p.” Differentiating, $H'(p) = \log_2\left(\frac{1-p}{p}\right)$, which vanishes at $p = 1/2$, giving $H(1/2) = 1$ bit. By Theorem 4 this is the global maximum, as $\log_2 2 = 1$.